

# Ekaterine Dadiani

Ph.D. candidate in Physics, Carnegie Mellon University  
PITTSBURGH, PA · [edadiani@andrew.cmu.edu](mailto:edadiani@andrew.cmu.edu)

## EDUCATION

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**Ph.D. Candidate in Physics – Carnegie Mellon University** **2023 – Present**  
Research topic: Studying LISA sources using DESI and LIGO/Virgo/KAGRA  
Pittsburgh, PA  
Advisors: Antonella Palmese, Tiziana Di Matteo

**M.Sc. in Physics – Carnegie Mellon University** **2021 – 2023**  
Thesis: Morphology of  $z \sim 2$  Dual AGN Host Galaxies in ASTRID Simulation  
Pittsburgh, PA  
Advisor: Tiziana Di Matteo

**B.Sc. in Physics – Ivane Javakhishvili Tbilisi State University** **2016 – 2020**  
GPA – 4.0 out of 4.0  
Tbilisi, Georgia  
Thesis: “Transient Flow and Magnetic Field Generation in 2 Temperature Degenerate Electron-Ion Gas due to Catastrophic Relaxation of Equilibrium”  
Advisor: Nana Shatashvili

## GRANTS & FELLOWSHIPS

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**NASA LISA Preparatory Science Grant** **2023–2026**  
PI: Antonella Palmese, Tiziana Di Matteo  
“Studying LISA sources using DESI and LIGO/Virgo/KAGRA”, \$687k.

**LSST Discovery Alliance Data Science Fellow** **2025–2027**  
Six one-week intensive sessions: Statistics & Bayesian Inference (U. Pittsburgh, May 2025); Image Processing for Wide-Field Surveys (Adler Planetarium, Sep 2025); Machine Learning for Astrophysics (SkAI Institute, Jun 2026); Scientific Communication & Data Visualization (UC San Diego, Aug 2026); two further sessions in 2026–27.

**Data Science Pathway of the CI Pathways Program** **2025**  
*NSF Award 2417789; Led by NCSA & PSC*  
A focused 10-week training program in data science, machine learning, and AI.

## COLLABORATIONS & MEMBERSHIPS

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Since 2023, I have actively contributed to two major international collaborations:

- **Dark Energy Spectroscopic Instrument (DESI)** ( $\sim 500$  members): Currently working on Project 249, “Searching for Massive Black Hole Binaries in DESI.”
- **ESA/NASA Laser Interferometer Space Antenna (LISA) Consortium** ( $\sim 1,000$  members): Member of the LISA Consortium within the Astrophysics Working Group at Carnegie Mellon University, focusing on modeling dual AGN host galaxies and predicting massive black hole binary events with electromagnetic counterparts.

## RESEARCH & WORK EXPERIENCE

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**GRADUATE RESEARCH ASSISTANT** **Jan 2024 – Present**  
**Carnegie Mellon University** **Pittsburgh, PA**

**Massive black hole binaries and dual AGN in DESI** (with Antonella Palmese)  
Leading the first systematic search for sub-parsec massive black hole binaries in DESI spectra, combining Doppler velocity offsets of broad emission lines with unsupervised deep learning (autoencoders trained on the full quasar sample) to separate orbital signatures from intrinsic AGN variability.  
Built the largest, uniformly selected census of kpc-scale dual AGN in DESI and analysed their host galaxy properties ([Dadiani et al. 2026](#)). Constructed matching mock DESI samples from the ASTRID

cosmological simulation and evolved the observed pairs forward to forecast LISA and PTA gravitational-wave detection rates (Chen, Zhou, Dadiani et al. 2026).

#### **Dual AGN host galaxies in cosmological simulations** (with Tiziana Di Matteo)

Measured the stellar kinematics and morphology of  $z \sim 2$  dual AGN hosts in ASTRID, showing that they are predominantly rotation-supported disks rather than merger-disturbed spheroids, and linked disk dynamics and gas inflow to black hole fueling (Dadiani et al. 2024).

#### **TEACHING ASSISTANT**

**Aug 2021 – Dec 2023**

**Carnegie Mellon University**

Pittsburgh, PA

Supported courses in Electronics 1, Physics I for Engineering Students, Physics II for Engineering Students, Matter and Interaction 1, and Matter and Interaction 2.

#### **SUMMER STUDENT, BE-ABP-HSC**

**Jul 2019 – Aug 2019**

**CERN**

Geneva, Switzerland

Derived closed-form analytical expressions for indirect space-charge wake functions in a circular beam pipe (supervisor: Nicolas Mounet); published in JACoW HB2021.

#### **VISITING SCHOLAR**

**Aug 2019 – Sep 2019**

**KU Leuven, Center for Mathematical Plasma Astrophysics**

Leuven, Belgium

Self-similar solutions for astrophysical MHD transients (supervisors: Andria Rogava, Stefaan Poedts); published in Adv. Space Res. 2022.

#### **ASSISTANT RESEARCHER**

**Oct 2018 – Oct 2019**

**Centre for Theoretical Astrophysics, Ilia State University**

Tbilisi, Georgia

Developed new classes of self-similar MHD flow solutions for coronal eruption models and refined their geometric and kinematic assumptions.

#### **FOUNDER AND CEO**

**2015 – 2017**

**TeddyBot and Quark**

Tbilisi, Georgia

Co-founded TeddyBot, an electronic childcare device. Won Seed Forum Georgia (2016) and was selected among the top 6 companies at Seed Forum Estonia. Press coverage.

Led Project Quark, a Sphero-based guide robot with waypoint navigation and route memory; secured funding offers from Georgian government organizations, including the Scientific Library of Georgia. Funding announcement.

#### **LEADERSHIP & OUTREACH**

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- **Organizing Committee, Astronomy on Tap Pittsburgh** (2025 – present).
- **Organizer, International Conference of Physics Students (ICPS)**, Tbilisi, Georgia (2024).
- **Local Jury & Organizing Committee, International Young Physicists' Tournament (IYPT)**, Tbilisi, Georgia (2019).

#### **PUBLICATIONS**

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Dadiani E., et al., “Cosmic Pairs: A DESI Census of Massive Black Hole Binaries” (2026) – In prep

Dadiani E., et al., “Cosmic Pairs: A DESI Census of Dual and Offset AGN as Precursors to Massive Black Hole Binaries” (2026) – Submitted to ApJ, <https://doi.org/10.48550/arXiv.2607.27390>

Chen N., Zhou Y., Dadiani E., et al., “Connecting Current and Future Dual Active Galactic Nucleus Searches to LISA and Pulsar Timing Array Gravitational-wave Detections”, ApJ 1008, 89 (2026) – <https://doi.org/10.3847/1538-4357/ae84ce>

Dadiani E., et al., “ $z \sim 2$  Dual AGN Host Galaxies are Disky: Stellar Kinematics in ASTRID Simulation,” (2024) – <https://doi.org/10.33232/001c.124451>

Rogava A., Poedts S., Dadiani E., “Self-Similarity for Astrophysical MHD Transients Revisited,” (2022) – <https://doi.org/10.1016/j.asr.2021.09.020>

Mounet N., Zannini C., Dadiani E., et al., “Closed Form Formulas of the Indirect Space Charge Wake

*Function for Axisymmetric Structures,” JACoW HB2021 (2022) – <https://cds.cern.ch/record/2841806/files/document.pdf?version=1>*

Saralidze E., Shatashvili N.L., Mahajan S.M., **Dadiani E.** “*Catastrophic formation of macro-scale flow and magnetic fields in the relativistic gas of binary systems*” – <http://dx.doi.org/10.1007/s10509-025-04456-1>

## CONFERENCES

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- Presented a talk at *16th International LISA Symposium*, College Park, Maryland (2026)
- Presented a talk at *The 8th Neighborhood Workshop*, Penn State (2026)
- Attended 2-week intensive program *The LISA School for Early-Career Scientists*, Ecole de Physique des Houches in the French Alps (2025)
- Presented a talk at *Cosmology on the Beach*, Playa del Carmen, Mexico (2024)
- Presented a talk at the *DESI Collaboration Meeting*, Marseille (2024)
- Presented a talk at the *First Pre-LISA Electromagnetic Mini-Workshop* (2023)
- Attended the *NanoGrav Workshop*, Pittsburgh (2023)
- Presented a talk at *The 5th Neighborhood Workshop*, Penn State (2023)
- Attended the *Second Michigan Cosmology Summer School* (2023)
- Presented a talk at the *Tbilisi State University VIII Faculty of Exact and Natural Science Students Conference* (2020)
- Presented a talk at the *International Conference of Physics Students (ICPS)*, Helsinki, Finland (2018)
- Presented a talk at the *Young Physicists Conference*, Tbilisi, Georgia (top prize, 2017)
- Presented a talk at the *8th Young Researchers Meeting*, Cagliari, Italy (2017)
- Presented a talk at the *7th Young Researchers Meeting*, Turin, Italy (2016)

## SELECTED AWARDS

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- *Finalist*, Intel International Science and Engineering Fair (ISEF), Phoenix, AZ (2016)
- *Winner*, Seed Forum Georgia (2016); top-6 company, Seed Forum Estonia (2016)
- *1st Place*, FabJam Georgia (2016); *1st Place*, “Leonardo Da Vinci” Competition (2016); *1st Place*, Georgian Young Physicists’ Tournament (2015); *1st Place*, International Young Naturalists’ Tournament (2015)
- *2nd Place*, Garage48 hackathon (2015); *3rd Place*, Millennium Innovative Challenge (2016), TSU Innovation Challenge (2016), and Ilia State University Hardware Hackathon (2015)
- Top-10 nationally, Georgian Educational Olympiad in Physics and Mathematics (2011 – 2015)

## RESEARCH INTERESTS

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Massive black hole physics and merger dynamics; Electromagnetic counterparts of gravitational wave events; Active Galactic Nuclei and dual AGN; Galaxy morphology and evolution; Dark Energy Spectroscopic Instrument; Synthetic observation pipelines and machine learning applications in astrophysics; Cosmological simulations; ...

## TECHNICAL SKILLS

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**Languages:** Python (primary), SQL, Bash, L<sup>A</sup>T<sub>E</sub>X

**Machine learning & data:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, Astropy; autoencoders and unsupervised anomaly detection; image processing for wide-field surveys

**Statistics:** Bayesian inference, MCMC, hierarchical models, time-series analysis, hypothesis testing on large catalogs

**Simulation & computing:** cosmological and N-body simulations (ASTRID), HDF5, HPC/Slurm, Git/GitHub, Jupyter